higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

16 August 2023 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used.

This question paper consists of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. All drawings must be neat and balanced.
5. Use only a blue or black pen.
6. Write neatly and legibly.

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QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.6) in the ANSWER BOOK.

1.1 The Boolean expression for a 3-input AND gate is …

A X = AB.
B X = ABC.
C X = A + B + C.
D X = AB + C.

1.2 How many bits does the 101011 binary number system have?

A 2 bits
B 4 bits
C 8 bits
D 6 bits

1.3 A full adder logic circuit will have …

A two inputs and one output.
B three inputs and three outputs.
C two inputs and two outputs.
D three inputs and two outputs.

1.4 How many data select lines are required for selecting eight inputs?

A 1
B 2
C 3
D 4

1.5 Which of the following options represent the synchronous control inputs in an SR flip-flop?

A Both S and R
B R
C Clock
D S

1.6 How many types of number systems are there?

A 1
B 2
C 3
D 4

(6 × 1) **[6]**

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QUESTION 2

2.1 What are *self-complementing codes*? (2)

2.2 Convert 0101011 Gray code to binary number system. (3)

2.3 Calculate the following in binary number system. Show all the conversions and calculations clearly:

2.3.1 $6{,}5_8 - 9{,}75_{10}$
Use the two's complement method and convert the final answer to the hexadecimal system. (6)

2.3.2 $4{,}75_{10} \times 15{,}5_8$ Convert the final answer to the octal system. (6)

2.3.3 $12{,}125_{10} \div 3{,}1_8$ Leave the answer in binary number system. (6)

[23]

QUESTION 3

3.1 Write down the simplest form of each Boolean expression below:

3.1.1 $A + AB$

3.1.2 $1 + A\bar{B}\bar{C}$

3.1.3 $A + \bar{A}B$

(3 × 1) (3)

3.2 Simplify the following logic expressions by making use of Boolean algebra:

3.2.1 $\bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC = F$ (7)

3.2.2 $\overline{\overline{(\bar{A}BC + A\bar{B}C)}\,\overline{(AB\bar{C} + ABC)}} = F$ (6)

3.2.3 $\overline{(A + B + \bar{C})(A + \bar{B} + C)(\bar{A} + B + C)(\bar{A} + \bar{B} + \bar{C})} = F$ (6)

3.3 Draw a logic circuit diagram of each Boolean expression below. Do not change or simplify the expression.

3.3.1 $\overline{\overline{A\bar{B}} + \overline{B\bar{C}} + \overline{AB\bar{C}}} = F$ (6)

3.3.2 $\overline{(AB \oplus \overline{AB}) \oplus \overline{A\bar{B}\bar{C}}} = F$ (6)

[34]

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QUESTION 4

4.1 Draw a neatly labelled diagram of a 4-bit twisted ring counter that uses JK flip-flops and draw a timing diagram for NINE clock pulses. (8)

4.2 Draw the circuit diagram of a module 5 counter and also give its truth table. (8)
[16]

QUESTION 5

5.1 Explain the difference between *encoder* and *decoder*. (4)

5.2 Explain the working principle of a successive approximation A/D converter. (7)

5.3 Draw a circuit diagram of a clocked RS flip-flop. Include the truth table. (10)
[21]

TOTAL: 100

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